CEGE 4160/5180 - Course Syllabus Applied Machine Learning for Civil, Environmental, and Geo- Engineers

Class: 9:45AM - 11:00AM MW, CivE 202

This course is designed for graduate students and senior undergraduate students who are interested in machine learning for civil engineering applications. The course offers an overview of models ranging from classical machine learning methods to state-of-the-art deep learning methods, and their applications in civil engineering problems. Specific topics include: regression, classification, and neural networks. All topics would have one or two example applications in the civil engineering context.

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1 Teaching Team

1.1 Instructors

Dr. Seongjin Choi

Email:	chois@umn.edu
Office:	CivE 162
Office Hours:	TBD

Dr. Choi received the B.S., M.S., and Ph.D. degree in the Department of Civil and Environmental Engineering from Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea, in 2015, 2017, and 2021, respectively. Dr. Choi's research interests are broad and interdisciplinary, encompassing Urban Mobility Data Analytics, Spatiotemporal Data Modeling, Deep Learning & Artificial Intelligence, and Connected Automated Vehicles (CAV) & Cooperative-ITS. He is particularly driven by the desire to optimize urban mobility and contribute to the development of a sustainable and efficient urban transportation system. His works involve utilizing data analytics to draw valuable insights from urban mobility data and applying cutting-edge AI technologies in the field of transportation.

1.2 Teaching Assistants

Mr. Lindong Liu

email:	liu03035@umn.edu
Office Hours:	TBD

1.3 Office Hours

- `https://choi-seongjin.github.io/contact/`

2 About the course

2.1 Course Focus

Civil, Environmental, and Geo-Engineering is rapidly becoming a data-rich field, with vast amounts of data generated from various sources such as structural sensors, traffic sensors, remote sensing, and massive simulation-based models. This course is designed for graduate students and senior undergraduate students who are eager to leverage machine learning techniques for applications in civil, environmental, and geo-engineering.

The course provides a comprehensive overview of machine learning models, spanning from classical methods to state-of-the-art deep learning approaches, tailored specifically for CEGE problems. Each week, specific topics will be addressed through a combination of theoretical lectures and practical examples. The course emphasizes hands-on learning, with one or two example applications in the civil, environmental, and geo-engineering context for each topic.

2.1.1 Ultimate Aim

By the end of this course, students will be able to understand basic concepts of machine learning, implement existing machine learning models, and analyze the results to solve complex, real-world problems in civil, environmental, and geo- engineering.

2.1.2 Foundational Aim (for building basic knowledge and understandings)

1. Learn the basic concepts of artificial intelligence, machine learning, and deep learning
2. Learn the basic data preprocessing and visualization
3. Gain proficiency in tools and programming languages commonly used in machine learning (Python, Numpy, Pandas, Pytorch)

2.1.3 Mediating Aim (For activities, assignments, and exams)

1. Implement machine learning algorithms on real-world datasets
2. Evaluate model performance, analyze the results, and refine models
3. Develop and document a machine learning pipeline

2.2 Course Prerequisites

- Math 1271, Math 1371, or equivalent (Calculus I)
- Math 1272, Math 1372, or equivalent (Calculus II)
- CEGE 3101, or equivalent (Computer Applications I)
- CEGE 3102, or equivalent (Uncertainty and Decision Analysis or basic statistics course)
- Ability to understand basic linear algebra, probability theory, and distribution, and write them into computer programs.

2.3 Text Books

There is **NO** official textbook required in this course.

While there is no official textbook, the instructor will share relevant reading materials as needed throughout the course. There are four optional textbooks for further reference, which are freely available online:

- *Probabilistic Machine Learning An Introduction* by Kevin Patrick Murphy[1]
- *Probabilistic Machine Learning: Advanced Topics* by Kevin Patrick Murphy[2]
- *Probabilistic Machine Learning for Civil Engineers* by James-A. Goulet[3]
- *Data-Driven Science and Engineering: Machine Learning, Dynamical Systems, and Control* by Steven L. Brunton and J. Nathan Kutz[4]

2.4 Course Web Page

The course web site will be used for messages and announcements, data transfer and code sharing, assignments, forums, questionnaires, and to browse your currents scores and standing in the course. The course web site is available through the University of Minnesota *Canvas* server. If you are unfamiliar with *Canvas* I recommend that you run through the University of Minnesota's "Canvas orientation" at

- Canvas: Getting Started for Students

To access the course web site go to

- `https://canvas.umn.edu/`

login, and go to the CEGE 4160/5180 site. Make it a habit to look-over this site at least a once a day.

2.5 Grading

The overall course grade will be based on five components: homework assignments, lab work, a midterm exam, a final project, and attendance. The approximate breakdown is as follows:

Homework	$15\% \pm 5\%$	3 individual assignments @ 50 pts per
Lab works	$15\% \pm 5\%$	5 in-class lab session assignments @ 30 pts per
Midterm Exam	$30\% \pm 5\%$	1 two-hour exam @ 300 pts
Final Project	$30\% \pm 5\%$	1 final project @ 300 pts (proposal: 100 pts, presentation: 100 pts, final report: 100 pts)
Attendance	$10\% \pm 5\%$	

[1] `https://probml.github.io/pml-book/book1.html`
[2] `https://probml.github.io/pml-book/book2.html`
[3] `https://profs.polymtl.ca/jagoulet/Site/Goulet_web_page_BOOK.html`
[4] `https://databookuw.com/`

2.6 Tentative Outline

The following is a tentative outline for the class. Reality can significantly deviate from this outline...

Week	Dates		Topic
0	-	1/22	Orientation
1	1/27	1/29	Review of Statistics, Linear Algebra, Optimization, Programming
2	2/03	2/05	Regression **- Lab01**
3	2/10	2/12	Regression **- HW01 :** Urban trees and canopy temperature
4	2/17	2/19	Classification **- Lab02**
5	2/24	2/26	Classification **- HW02 :** Predicting travel mode choice behaviors
6	3/03	3/05	Neural Networks - Fundamentals **- Lab03**
7	3/10	3/12	*Spring Break*
8	3/17	3/19	Neural Networks - Structures **- Project Proposal**
9	3/24	3/26	Regression and Classification using Neural Networks **- Lab04**
10	3/31	4/02	Object Detection **- Lab05**
11	4/07	4/09	Semantic Segmentation **- HW03** : **- Choice 1** - Concrete crack detection **- Choice 2** - Visual scene understanding for self-driving cars
12	4/14	4/16	TBD - Spatiotemporal prediction - Physics-informed machine learning - Generative AI
13	4/21	4/23	TBD (Guest Lectures)
14	4/28	4/30	**Final Project Presentation**
15	5/05	-	Summary - **Final Project Report**